# MUHAMMAD DAFFA ATHA RIZQ



## Data Science & Machine Learning Specialist

heading toward AI Engineer & AI Agent Developer

Bekasi, West Java

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## PROFESSIONAL SUMMARY

B.Sc. Informatics graduate of Gunadarma University (GPA 3.61) focused on Data Science & Machine Learning: building end-to-end ML pipelines based on CRISP-DM, processing datasets of more than 2.9 million transactions, and creating interactive web applications. Backed by field experience in logistics administration and database management. Currently deepening AI Engineering through PrivateAI, a local & offline AI agent built on the Human-in-the-Loop principle. Open to job opportunities.

## SKILLS

**Machine Learning & AI:** XGBoost, Scikit-Learn, Sentence-BERT, TruncatedSVD, Multinomial Naive Bayes, Deep Learning, Feature Engineering, Time-Series Forecasting

**Local AI Agent & LLM:** Ollama, Qwen2.5, Gradio, SQLite, Pytest, safety gate & human approval design

**Data Analytics:** Pandas, NumPy, Matplotlib, Seaborn, CRISP-DM, EDA

**Programming & Web:** Python, SQL, JavaScript, HTML5, CSS3, Streamlit, Flask, Bootstrap

**Tools & Database:** MySQL, phpMyAdmin, Git, GitHub, Google Colab

**Operations:** Fleet Management, Financial & Contract Reporting, Crisis Management

## PROJECTS

### PrivateAI v1.8 — Local & Offline AI Agent (MVP)

AI Agent Engineering

*Python · Ollama · Qwen2.5 1.5B · Gradio · SQLite · Pandas · Pytest*

- Designed an AI agent that runs entirely on a laptop with no internet (8 GB RAM, no GPU): the user types an ordinary task, the agent writes and runs the code, then waits for human approval before the result is final.
- Applied the "Small LLM + Strong Engineering" principle: the LLM only understands intent and writes code, while security, permissions, validation, and auditing are handled by ordinary code — pre-execution safety checks, a restricted workspace, automatic repair (max 3×), and an audit log.
- 3,226 lines of application code with 141 passing tests across 22 test files (mock LLM with a real backend) and a 13-task benchmark, Gradio GUI, SQLite memory, and seven layers of defense. Honestly scoped as a personal-use MVP (not an OS-level sandbox).

### E-Commerce Product Demand Forecasting

Machine Learning & Data Analytics

*Python · XGBoost · Scikit-Learn · Pandas · Seaborn*

- Modeled monthly product demand estimates (time-series) based on CRISP-DM on more than 2.9 million rows of Kaggle transaction data.
- XGBoost Regressor was the best model (RMSE 0.626; MAE 0.293; $R^2$ 0.630), outperforming Stacking Ensemble and Random Forest.
- Built an interactive analytics dashboard with 18 business charts to read demand trends from data.

### CineMatch AI — Hybrid Movie Recommendation System

AI System & Web Application

*Python · Streamlit · Sentence-BERT · TruncatedSVD · Multinomial Naive Bayes*

- Built a hybrid movie recommender (TruncatedSVD + Sentence-BERT + Multinomial Naive Bayes) with a Strict Mode and proportional quota allocation to boost the visibility of local Indonesian films.
- Evaluated with Precision/Recall and tested through Blackbox Testing and User Acceptance Testing (UAT).

## WORK EXPERIENCE

**Logistics Vendor Company Admin** Oct 2025 – Jul 2026 (Part Time)
CV Barokah Bersama Logistik, Cibitung, Bekasi Regency

- Managed the company's operational database and data entry accurately.
- Prepared and analyzed operational financial reports and partnership contracts.
- Handled logistics, scheduling, and daily coordination of vehicle units (Fleet Management).
- Responded to emergency fleet crises in the field.

**Contract Logistics Data & Billing Staff** Jun 2026 (Project-Based)
PT Sinar Sosro (via Shipper.id)

- Managed the recording of shipment transaction data.
- Issued Proforma Invoices and Delivery Notes.
- Calculated gross-up tax on billing.

## EDUCATION

**B.Sc. Informatics (S.Kom.)** | Gunadarma University 2022 – 2026

- GPA **3.61 / 4.00**
- Thesis: Implementation of Machine Learning & Deep Learning in the CineMatch AI Recommendation System & E-Commerce Demand Forecasting.

**SMAN 9 Tambun Selatan (Senior High School)** | Bekasi Regency 2019 – 2022

**SMPN 12 Tambun Selatan (Junior High School)** | Bekasi Regency 2016 – 2019

## CERTIFICATIONS

**Databases:** Advanced Database Systems (Jun 2026); Fundamentals of Database Systems (Sep 2025); Oracle for Intermediate (Aug 2025); Oracle for Beginner (Aug 2024); Fundamental DBMS (Aug 2023)

**Web & Programming:** Go-Lang for Intermediate (Feb 2025); Building Website using HTML 5 (Jan 2025); Go-Lang for Beginner (Feb 2024); Fundamental Web Programming (Feb 2023)

**Other:** Aptitude Test (Jul 2026); GDSC Gunadarma University Info Session — Google Developer Student Clubs Gunadarma (2022)

*Certificates issued by Gunadarma University unless stated otherwise.*